Spam Detection Classification Report

1.Introduction

The goal of this project is to create a supervised machine learning model to categorize SMS messages as either ham or spam. Ham messages are typical personal communications, whereas spam messages are unsolicited promotional or fraudulent messages. The objective is to develop a model that can recognize spam messages on its own.

2.Data Set

The project uses the SMS Spam Collection Dataset, which contains labelled SMS messages.

Dataset details:

- Total messages: **5572**
- Classes:
  - **Ham (non-spam)**
  - **Spam**

Each record contains:

- Label (spam or ham)
- Message text

3. Data Pre-processing

The following pre-processing steps were performed using **pandas**:

- Loaded dataset

- Converted labels to numeric values
    - Ham → 0
    - Spam → 1
- Created message length feature
- Converted text messages to numerical features using TF-IDF

4. Data Visualization

Several plots were created to understand the dataset:

- Bar chart showing spam vs ham distribution
- Histogram of message length
- Boxplot of message labels
- Scatter plot of message length vs label

These visualizations help understand how spam and ham messages differ.

5. Machine Learning Models

Two models were implemented using **scikit-learn**:

1. Logistic Regression
2. Random Forest Classifier

The dataset was split into training (80%) and testing (20%) sets.

6. Model Evaluation

The models were evaluated using the following metrics:

- Accuracy

- Precision
- Recall
- F1 Score
- ROC-AUC

Cross-validation with 5 folds was also performed to validate model performance.

7. Results

Both models performed well in classifying SMS messages. Logistic Regression achieved strong performance for text classification due to its effectiveness with TF-IDF features. Random Forest also performed well but required more computational resources.

8. Conclusion

This project successfully demonstrates how machine learning can be used to detect spam messages. The trained model can classify SMS messages as spam or ham with high accuracy. Such systems can help filter unwanted messages and improve user communication experience.